As part of the energy efficiency team, your task is to analyse the provided data set and answer the following questions. These questions will guide your exploration of the data and help you uncover key insights.

Questions to Answer

By linking different sensor trends, analyse patterns and abnormalities. Use sensor data for the evidence.

1) 08:00 to 11:30, as the number of students increases, which two other measurements also consistently go up?
2) At 15:30, one sensor gives a reading of 9999. Why is it scientifically impossible for this to be a real reading caused by the students in the room?
3) A "Smart Building" should turn its lights down when the sun is bright. Between 09:00 and 12:00, does the data show the building successfully saving energy?
4) Identify the slots where the room was most likely uncomfortable for students. There are 3 considerations of unusual data readings for the time frames and adjacent sensor logs.
    a) Identify the specific 30-minute window where the room was most likely to be uncomfortable for studying. Which sensors provide the "evidence" that it was too hot or the air was too poor?
    b) At 12:00, the temperature sensor shows an ERROR. Based on the trend of the numbers immediately before and after that time, what do you think the actual temperature was?
    c) There is a discrepancy in standard lightning patterns for the times of day, identify what you feel may be the cause
5) Identify the 4 specific data failures in the table and propose a "Corrected Value" using logic (interpolation or trend matching).

Dashboards Design

Create a simple paper based dashboard that demonstrates the data as bar grahs and charts which can provide trends